Abstract:

Basic models like OpenPose and MMpose struggle to predict keypoints even under minor occlusion. To address this, I fine-tuned an HRNet-W18 model specifically for occluded keypoints. I created train, validation, and test datasets focused on a left-side viewpoint of people performing barbell back squats. The scope of the project was narrowed to improve predictions for the left shoulder and left elbow.

For all datasets, I gathered videos featuring varying degrees of clothing, body types, and genders. For the training set, I ran OpenPose on each video and manually corrected the keypoints of interest, assigning confidence scores to each annotation. For the validation set, I manually annotated the left shoulder, left elbow, left hip, and left knee using CVAT and created custom bounding boxes. After modifying the MMPose configuration to HRNet-W18 and the custom datasets, I trained the model. The model was unable to produce accurate predictions, largely due to narrow scope and limited size of the dataset, which prevented meaningful generalization.

Introduction:

I wanted to help people improve their lifting form in the gym. About four years ago, I suffered a back injury caused by improper lifting technique, which motivated me to explore ways to prevent similar injuries in others. I envisioned an application that could analyze user-submitted videos and provide rule-based feedback on lifting form.

To achieve this, accurate keypoint predictions are required. However, during barbell lifts, shoulders and elbows are frequently occluded by the barbell plates, making reliable rule-based analysis difficult. Existing pose estimation models struggle to predict occluded keypoints, particularly in these scenarios. As a result, I set out to fine-tune a model specifically to address occluded shoulder and elbow prediction. I hypothesized that if a model could accurately infer an occluded keypoint from a single viewpoint, it could generalize to other viewpoints given sufficient data.

Related Research:

Occlusion is a major challenge in pose estimation. In sports settings, ideal camera angles are rare due to other athletes, equipment, padding, and environmental constraints. Current pose estimation models operate on a frame-by-frame basis and do not incorporate temporal information from past or future frames. As a result, predictions are often inaccurate when keypoints are partially or fully occluded.

Methods/Approach

Dataset:

The dataset was split into training, validation, and test sets.

Train set:

The training set consisted of approximately 4,000 frames extracted from 10 videos of individuals performing barbell back squats from a left-side viewpoint. Camera angles varied slightly between videos (higher, lower, or shifted left/right), and each video contained 3–6 repetitions. The dataset included individuals of different genders, body types, and clothing layers.

Each video was first processed using OpenPose. I then manually annotated knees, hips, shoulders, and elbows. Shoulder and elbow annotations were frequently required due to occlusion. Confidence scores were assigned as follows: 0.95 (clearly visible), 0.75 (very good estimate), 0.5 (reasonable estimate), 0.25 (weak estimate), and 0 (impossible to estimate). The annotations were converted to COCO format, and unused keypoints were removed.

Validation set:

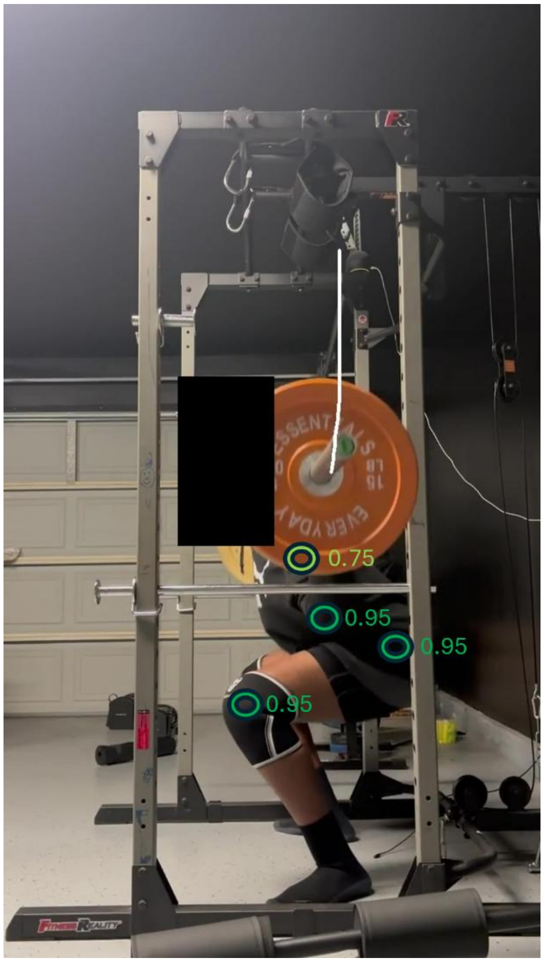
The validation set consisted of approximately 170 frames extracted from 13 entirely new videos. Fifteen frames were randomly sampled from each video. Using CVAT, I manually annotated the left shoulder, left elbow, left hip, and left knee, along with a custom bounding box. The same confidence scoring rules used for the training set were applied.

Test set:

The test set consisted of approximately 300 frames extracted from 20 additional videos not included in the training or validation sets. It was constructed using the same methodology as the validation set.

Visual:

To illustrate the annotation procedure, representative frames were taken from the validation set and annotated using color-coded markers. These visualizations reflect the same annotation criteria applied across the training, validation, and test sets. They demonstrate the camera angle and lift used in dataset creation, the confidence assignment criteria, and the specific keypoints that were annotated.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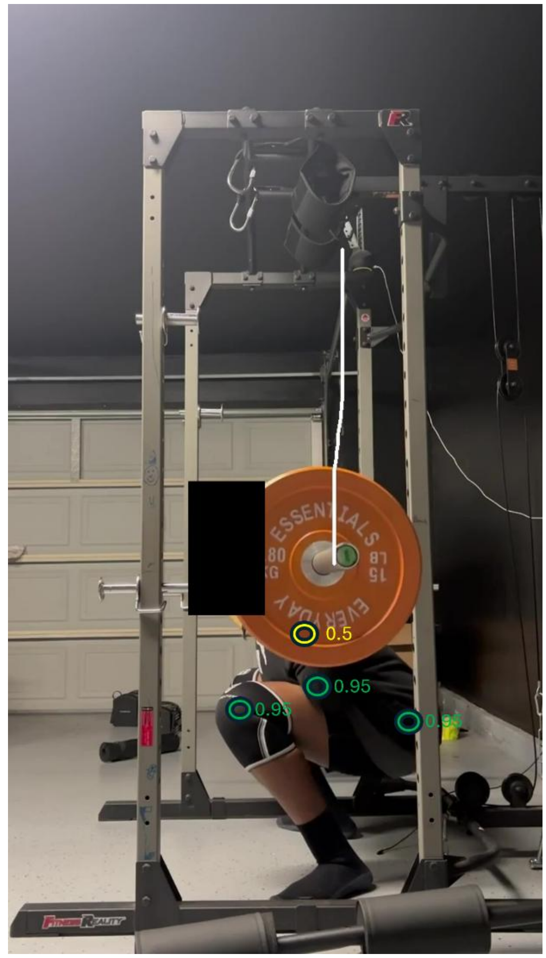
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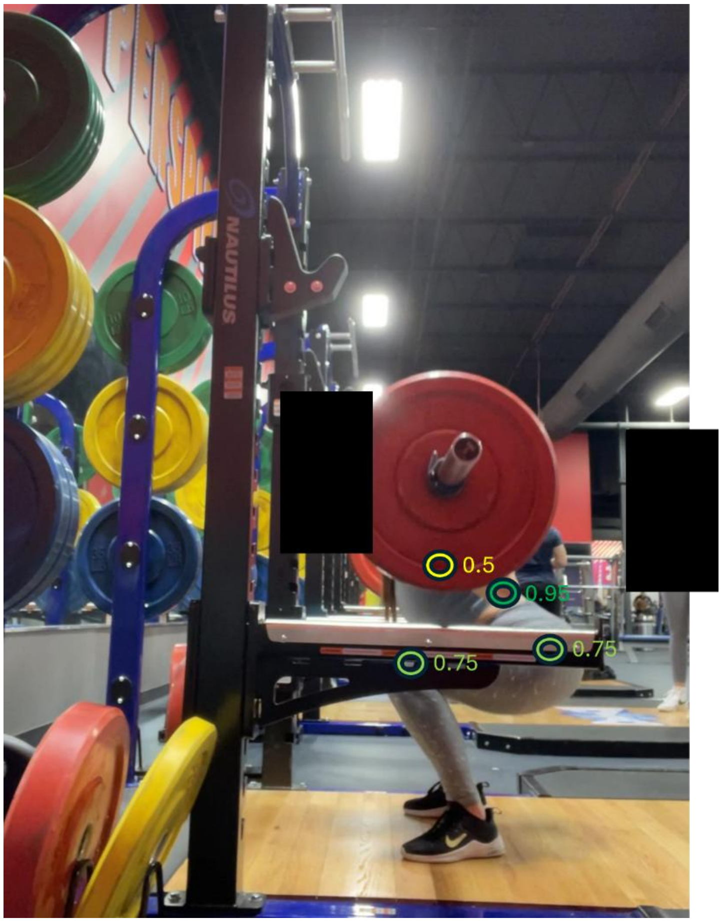


Model

Base Model:

HRNet-W18 backbone was selected due to its relatively low computational costs, making it feasible to train on limited hardware.

Key Configuration Adjustments:

1. Keypoint Definition Change
    - Reduced from the standard COCO 17 keypoints to a custom 4-keypoint schema.
    - Updated num_output_channels, num_joints, dataset annotation loaders, and heatmap head output dimensions
2. Output Channels
    - Modified the final layer to output four heatmaps corresponding to keypoints of interest.
3. Pretrained Weights
    - Initialized using ImageNet-trained HRNet-W18 weights.
4. Data Augmentation Changes
    - Disabled random flip, scale, and rotation to maintain consistency with bounding boxes.
5. Learning Rate Schedule
    - Set to 1e-4
6. Training Duration
    - Trained for 10 epochs (no meaningful learning occurred after epoch 1–2).

Augmentation:

To mimic real-world situations (camera angle, lighting, noise), synthetic augmentation produced a final training set of approximately 10,000 frames:

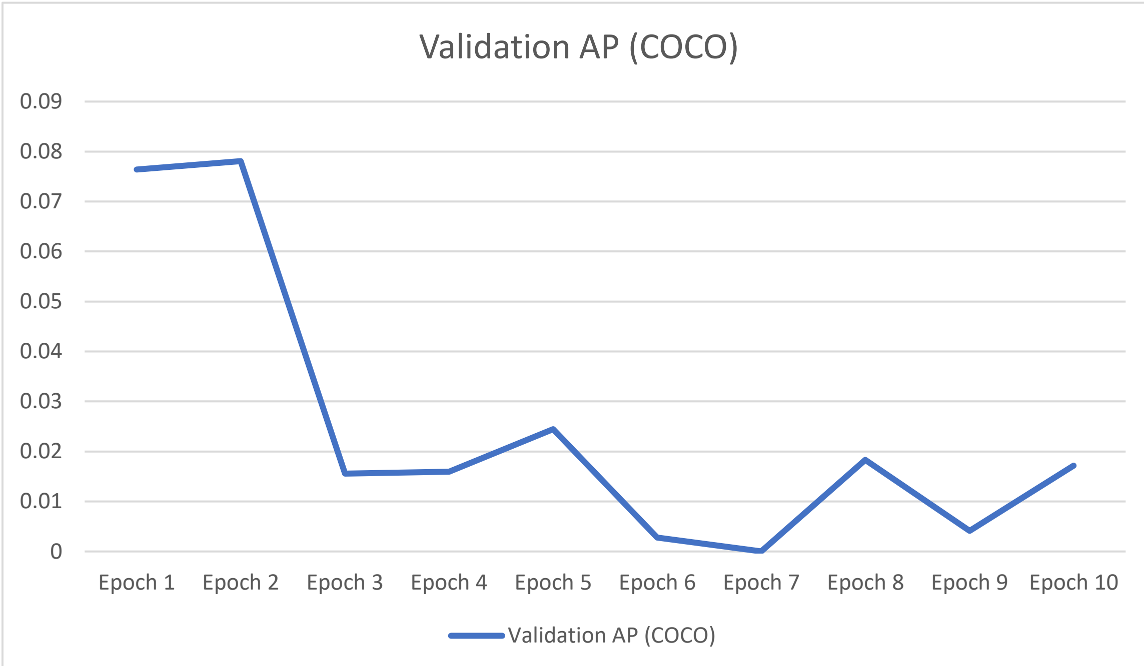
- 4,000 original
- 2,000 with minor synthetic occlusion
- 2,000 with minor lighting/transform effects
- 2,000 with minor Gaussian noise

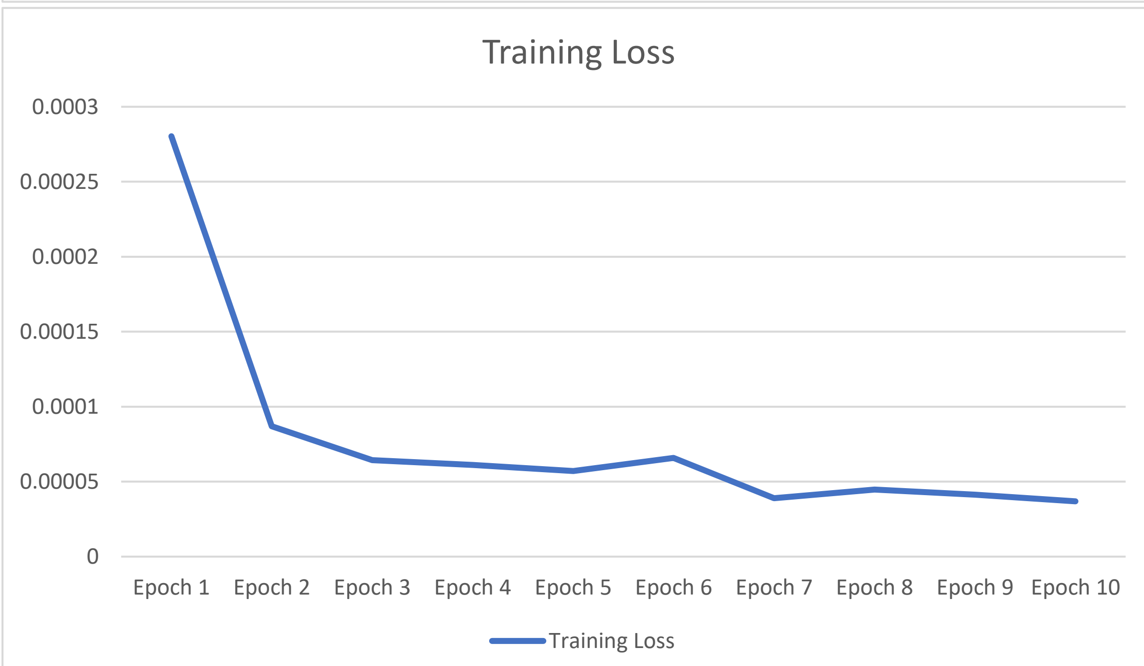
All effects were minor to avoid confusing the model. Bounding boxes were recalculated after each augmentation.

Results (quantitative):

Epoch	Training Loss	Validation AP (COCO)	Validation AP@50	Validation AP@75	Validation AR (COCO)	Best AP	Best Epoch
1	0.000280333	0.076396	0.188969	0.027439	0.142581	0.0764	1
2	8.69444E-05	0.078087	0.264558	0.042717	0.196129	0.0781	2
3	6.43889E-05	0.01558	0.036139	0.009901	0.016129	0.0781	2
4	6.10556E-05	0.015977	0.05599	0	0.027742	0.0781	2
5	0.000057	0.024456	0.076306	0.017602	0.052903	0.0781	2
6	6.58889E-05	0.002755	0.012788	0	0.016129	0.0781	2
7	3.90556E-05	0.000018	0.00018	0	0.000645	0.0781	2
8	4.48333E-05	0.018329	0.063493	0	0.024516	0.0781	2
9	4.13889E-05	0.00414	0.011341	0	0.005161	0.0781	2
10	3.68889E-05	0.01717	0.073281	0.009901	0.029677	0.0781	2

Validation AP and training loss across training epochs:





Results summary:

Average Precision (AP) measures keypoint localization accuracy across confidence thresholds, while Average Recall (AR) reflects the model's ability to detect keypoints regardless of precision. Across all training runs, including unrecorded experiments, the model exhibited very low training loss from the first epoch and effectively stopped learning immediately. The best average precision (AP) was captured within the first two epochs and did not improve afterward. This behavior was consistent across both augmented and non-augmented datasets. Because the model failed to learn beyond the first two epochs, further ablations were not pursued as results showed data limitations, not optimization issues.

Visualization:

Qualitative inspection of model outputs revealed limited structure in the predictions. While predicted keypoints were often placed on or near the subject's body, their positions varied inconsistently across frames and did not follow anatomically meaningful patterns. This behavior was observed for both occluded and non-occluded keypoints, indicating a failure to learn robust spatial representations.

Limitations:

The primary limitation of this work is the size and scope of the dataset. Even with augmentation, 4,000 original frames are insufficient for HRNet-W18 to learn robust representations. While restricting the viewpoint narrowed the problem space, limiting the task to only four keypoints reduced the model's ability to generalize across different human anatomies, lighting, and angles. Additionally, the annotations were manually generated and inherently noisy despite consistent confidence rules. Due to the model's early failure, confidence-based weighting was left for future work. Only HRNet-W18 was evaluated, but based on the limitations, similar results would likely be observed with other models.

Future Work:

Because of the primary limitations and the inability to determine which factors inhibited learning, it is unclear whether the model failed to generalize due to architectural or dataset limitations. Thus, future work should begin with the creation of a synthetic dataset in a 3D environment such as Blender. This allows for perfect keypoint annotation, controlled dataset size and scope, systematic occlusion, and control over pose and camera angles. After generating the dataset, HRNet-W32 should be evaluated to determine which factors affect model performance. After identifying limitations, fine-tuning can be revisited along with methods such as temporal smoothing, multi-view fusion, and the introduction of larger real-world datasets.